Separation of Variables

Exam code: 9MA0

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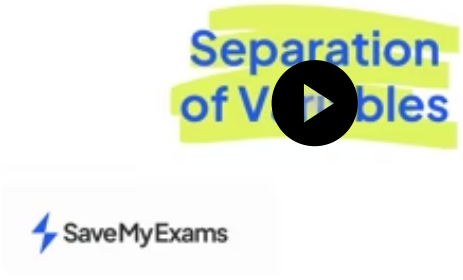
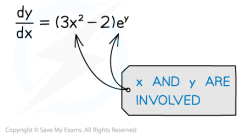
Help me understand



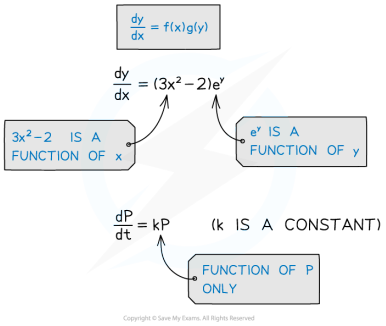
Did this video help you? Yes No

Separation of variables

What does it mean for a differential equation to be separable?



- Many differential equations used in modelling have two variables involved (ie **x** and **y**)
- If there is a product of functions in different variables, the differential equation is separable
  - ie **dy/dx = f(x) × g(y)**



Stuck on something?

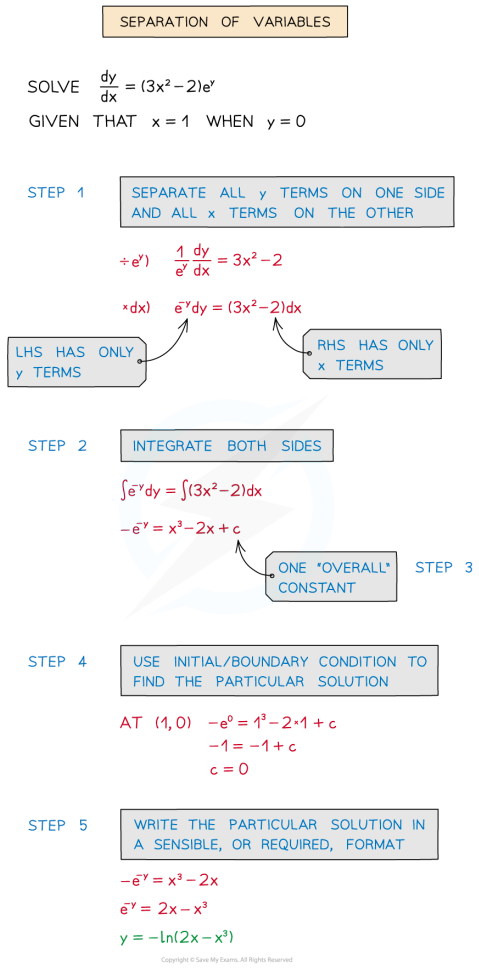
Ask questions and get help with exactly what you're revising.

Got it

thought of as **dy/dx= f × g(y)**

- where f(x) = 1

How do I solve a differential equation using separation of variables?



- STEP 1: Separate all y terms on one side and all x terms on the other side
- STEP 2: Integrate both sides
- STEP 3: Include one "overall" constant of integration
- STEP 4: Use the initial or boundary condition to find the particular solution
- STEP 5: Write the particular solution in sensible, or required, format

Worked Example

the differential equation $\frac{dA}{dr} = \frac{4\pi r + 2}{A}$

$$\frac{dA}{dr} = \frac{4\pi r + 2}{A}$$

g(r)

f(A)

STEP 1 SEPARATE VARIABLES (A AND r)

$A\,dA = (4\pi r + 2)\,dr$

STEP 2 INTEGRATE BOTH SIDES

$\int A\,dA = \int (4\pi r + 2)\,dr$

$\frac{A^2}{2} = 2\pi r^2 + 2r + c$

GENERAL SOLUTION

STEP 3 ONE "OVERALL" CONSTANT

STEP 4 USE THE BOUNDARY CONDITION

$20\pi + 4 = 4\pi \times 2^2 + 4 \times 2 + c$

$20\pi + 4 = 16\pi + 8 + c$

$c = 4\pi - 4$

STEP 5 WRITE THE PARTICULAR SOLUTION IN A SENSIBLE, OR REQUIRED, FORMAT

$A^2 = 4\pi r^2 + 4r + 4\pi - 4$

$A^2 = 4\pi(r^2 + 1) + 4(r - 1)$

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